# MATH 110 — Week 5

## Homework 5 — Solutions

**HW5 P1(a).**

**Idea.** The two evaluations force factors $x$ and $x-1$; linear interpolation gives every output.

1. Evaluation gives $T(ap+q) = (ap(0)+q(0), ap(1)+q(1)) = aTp+Tq$, so $T$ is linear (the additional exam ask). Also $Tp = 0$ iff $p(0) = p(1) = 0$. By the factor argument proved in Week 4, $p = x(x-1)q$ with $q \in \mathcal{P}_1(\mathbb{R})$. Conversely every such polynomial vanishes at both points.
2. Thus $\ker T = \operatorname{span}(x(x-1), x^2(x-1))$. The two polynomials are independent by their different degrees, so they form a basis.
3. For arbitrary $(a,b) \in \mathbb{R}^2$, the polynomial $p = a + (b-a)x$ has $Tp = (a,b)$. Hence $T$ is onto.

**HW5 P1(b,c).**

**Idea.** The induced map records the two evaluations; cosets of $1, x$ already produce an output basis.

1. Let $K = \ker T$. The induced isomorphism is $\overline{T}(p+K) = (p(0), p(1))$. Its inverse sends $(a,b)$ to $a + (b-a)x + K$, by part (a).
2. The cosets $1+K, x+K$ map to $(1,1), (0,1)$, an independent spanning list of $\mathbb{R}^2$. An isomorphism preserves bases, so these two cosets are a quotient basis.
3. Equivalently, every $p+K$ equals $p(0)(1+K) + (p(1)-p(0))(x+K)$, and evaluations force these coefficients uniquely.

**HW5 P2.**

**Idea.** The containment $U \subseteq \ker T$ ensures changing the representative does not change its image under $T$.

1. If $v+U = w+U$, then $v-w \in U \subseteq \ker T$, so $Tv = Tw$. Thus $\widetilde{T}(v+U) = Tv$ is well-defined.
2. $\widetilde{T}(a(v+U)+(w+U)) = T(av+w) = aTv+Tw$, proving linearity. Every quotient element is some $v+U$, so the prescribed formula also proves uniqueness.
3. A coset $v+U$ lies in the kernel iff $Tv = 0$, or $v \in \ker T$. Thus $\ker \widetilde{T} = \{v+U : v \in \ker T\} = (\ker T)/U$.

**HW5 P3.**

**Idea.** A single nonzero output can be scaled to every field element.

1. If $\varphi$ is not zero, choose $v$ with $a = \varphi(v) \neq 0$.
2. For each $b \in F$, $\varphi((b/a)v) = (b/a)\varphi(v) = b$. Hence $\varphi$ is surjective.

**HW5 P4.**

**Idea.** Dual basis values isolate one coefficient at a time.

1. Write $v = \sum_j a_j v_j$. Applying $\varphi_k$ gives $\varphi_k(v) = \sum_j a_j \delta_{kj} = a_k$, proving part (a).
2. For part (b), evaluate $\sum_j \psi(v_j)\varphi_j$ at $v_k$. The result is $\sum_j \psi(v_j)\delta_{jk} = \psi(v_k)$.
3. The two linear functionals agree on a basis, hence everywhere by Week 3 Proposition 21.

**HW5 P5.**

**Idea.** Entries of the original matrix are evaluations by the codomain dual basis.

1. Let $B = (v_1, \dots, v_n)$ and $C = (w_1, \dots, w_m)$, with dual bases $\varphi_j$ and $\psi_i$. Since $Tv_j = \sum_i a_{ij} w_i$, we have $(T'\psi_i)(v_j) = \psi_i(Tv_j) = a_{ij}$.
2. The functional expansion gives $T'\psi_i = \sum_j a_{ij}\varphi_j$, so entry $(j, i)$ of its matrix is $a_{ij}$.
3. This is exactly the transpose matrix $A^{\mathsf{T}}$.

**HW5 P6.**

**Idea.** Write the vanishing conditions on the two generators.

1. A row $(a, b, c)$ lies in $U^0$ iff $a + b = 0$ and $b + c = 0$, so it equals $b(-1, 1, -1)$. Thus $U^0 = \operatorname{span}((-1, 1, -1))$ over $\mathbb{R}$.
2. The generators of $U$ are independent because the first coordinate of a zero relation forces the first coefficient zero, and the third coordinate forces the second zero. Hence $\dim U = 2$.
3. The annihilator generator is nonzero, so $\dim U^0 = 1$. Therefore $\dim U + \dim U^0 = 2 + 1 = 3$.

**HW5 P7 (optional).**

**Idea.** Separate a nonzero vector by one functional, then use finite-dimensional equality of dimensions.

1. For fixed $v$, evaluation $j_V(v)$ is linear in the functional; the identity $j_V(av + w)(\varphi) = a\varphi(v) + \varphi(w)$ also makes $j_V$ linear.
2. If $v \neq 0$, extend $v$ to a basis and define $\varphi$ to be 1 on $v$ and zero on every other basis vector. Finite basis expansions define a linear functional; arbitrary-dimensional basis extension uses the cited choice result.
3. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$, proving zero kernel and injectivity.
4. If $V$ is finite-dimensional, $\dim V'' = \dim V' = \dim V$. The equal-dimensional injectivity criterion gives an isomorphism.

**Exam practice, Question 2(a,b).**

**Idea.** The dual map precomposes a functional with the original map.

1. Define $T^*(\varphi) = \varphi \circ T$ for $\varphi \in W^*$. It lies in $V^*$ since composition of linear maps is linear.
2. For $a \in \mathbb{R}$ and $\varphi, \psi \in W^*$, $T^*(a\varphi + \psi)(v) = a\varphi(Tv) + \psi(Tv) = (aT^*\varphi + T^*\psi)(v)$ for all $v$. Thus $T^*$ is linear.
3. For $\eta \in Z^*$, $(S \circ T)^*\eta = \eta \circ S \circ T = T^*(S^*\eta)$ by associativity, proving the product identity.

**Exam practice, Question 2(c).**

**Idea.** Surjectivity makes vanishing on all $Tv$ the same as vanishing on all of $W$.

1. If $T^*\varphi = 0$, then $\varphi(Tv) = 0$ for all $v \in V$.
2. For each $w \in W$, choose $v$ with $Tv = w$. Then $\varphi(w) = 0$, so $\varphi$ is the zero functional.
3. Thus $\ker T^* = \{0\}$, and the zero-kernel criterion proves injectivity.

**Exam practice, Question 2(d).**

**Idea.** Compose the given scalar functional with the map.

1. $(T^*\ell)(x, y) = \ell(T(x, y)) = 3(2x - y) = 6x - 3y$.
2. Thus $T^*\ell$ is the functional represented by row $(6, -3)$.

**Exam review, Problem 1(a).**

**Idea.** The three values determine the three coefficients of a function in $V$; solving these equations gives the functionals that return those coefficients.

1. Evaluation and derivative evaluation are linear by elementary calculus. For $f = a + b\sin x + c\sin 2x$, the values are

$$\begin{array}{c|ccc} & 1 & \sin x & \sin 2x \\ \hline \delta_0 & 1 & 0 & 0 \\ \delta_{\pi/2} & 1 & 1 & 0 \\ \delta_0' & 0 & 1 & 2 \end{array} \qquad \begin{pmatrix} \delta_0(f) \\ \delta_{\pi/2}(f) \\ \delta_0'(f) \end{pmatrix} = \begin{pmatrix} a \\ a+b \\ b+2c \end{pmatrix}.$$

2. If $f$ is the zero function, these values force $a = 0$, then $b = 0$, then $c = 0$. Hence $(1, \sin x, \sin 2x)$ is a basis of $V$, and $\dim V' = 3$.
3. If $\alpha\delta_0 + \beta\delta_{\pi/2} + \gamma\delta_0' = 0$, evaluating at $\sin 2x$ gives $2\gamma = 0$, at $\sin x$ gives $\beta + \gamma = 0$, and at 1 gives $\alpha + \beta = 0$. Thus all three coefficients vanish. Three independent functionals in dimension 3 form a basis.

**Exam review, Problem 1(b,c).**

**Idea.** Solve the displayed value equations for the coefficient functionals, then integrate each original basis function.

1. The solutions are $a = \delta_0(f)$, $b = \delta_{\pi/2}(f) - \delta_0(f)$ and $c = (\delta_0'(f) - \delta_{\pi/2}(f) + \delta_0(f))/2$. Therefore

$$\varphi_1 = \delta_0, \qquad \varphi_2 = \delta_{\pi/2} - \delta_0, \qquad \varphi_3 = \tfrac{1}{2}(\delta_0' - \delta_{\pi/2} + \delta_0).$$

   These return $a, b, c$, so their values on the original basis are the required delta values.
2. $L(1) = 1$, $L(\sin x) = 2/\pi$, and $L(\sin 2x) = 0$, by $\int_0^\pi 1 = \pi$, $[-\cos x]_0^\pi = 2$, and $[-\cos 2x/2]_0^\pi = 0$.
3. The functional expansion gives $L = \varphi_1 + (2/\pi)\varphi_2 + 0\varphi_3 = (1 - 2/\pi)\delta_0 + (2/\pi)\delta_{\pi/2} + 0\delta_0'$.

**Exam review, Problem 2(a).**

**Idea.** The dual of inclusion restricts values; the dual of the quotient precomposes a functional with $\pi$. Apply Proposition 50 with the exact kernel and range of each original map.

1. $i'\varphi = \varphi|_U$. It is zero exactly when $\varphi(U) = 0$, so $\ker i' = U^0$. Every functional on $U$ extends to $V$ by the extension tool, giving $\operatorname{range} i' = U'$.
2. The quotient map is onto with kernel $U$. Thus $\ker \pi' = (\operatorname{range}\pi)^0 = (V/U)^0 = \{0\}$, where the annihilator is taken in $(V/U)'$.
3. Also $\operatorname{range}\pi' = (\ker\pi)^0 = U^0 \subseteq V'$. This gives all four spaces with their ambient duals specified.

**Exam review, Problem 2(b).**

**Idea.** Two extensions of the same functional on $U$ differ by an element of $U^0 = \operatorname{range}\pi'$, so their cosets agree. This makes the extension construction independent of its choices.

1. For $\eta \in U'$, choose an extension $\widetilde{\eta} \in V'$ and set $F(\eta) = \widetilde{\eta} + U^0$. If two extensions are chosen, their difference vanishes on $U$, so they give the same coset. Thus $F : U' \to V'/U^0$ is well-defined.
2. If $\widetilde{\eta}, \widetilde{\theta}$ extend $\eta, \theta$, then $a\widetilde{\eta} + \widetilde{\theta}$ extends $a\eta + \theta$. Hence $F(a\eta + \theta) = aF(\eta) + F(\theta)$, proving linearity.
3. If $F(\eta) = U^0$, its extension vanishes on $U$, so $\eta = 0$; $F$ is injective. For any $\varphi + U^0$, take $\eta = \varphi|_U$; then $\varphi$ is an extension and $F(\eta) = \varphi + U^0$, proving surjectivity.
4. Proposition 29 from Week 4 makes $F$ an isomorphism. Part (a) identifies $U^0 = \operatorname{range}\pi'$, so its target is the required quotient. Its inverse sends $\varphi + U^0$ to $\varphi|_U$.

**Exam review, Problem 8, dual-map assertion.**

**Idea.** Injectivity of $T$ guarantees surjectivity of $T'$, not injectivity. A proper inclusion lets a nonzero functional vanish on the entire range.

1. Take $T : \mathbb{R} \to \mathbb{R}^2$, $T(t) = (t, 0)$. It is linear and injective since $T(t) = 0$ implies $t = 0$.
2. A row $\varphi_{a,b}(x, y) = ax + by$ gives $(T'\varphi_{a,b})(t) = at$. Thus every row $(0, b)$ is in $\ker T'$, including the nonzero functional $(x, y) \mapsto y$. The asserted implication is false.